

Problems: Categorical and Numeric Variables

Q1.

Consider the data: [red, blue, blue, red, green, grey]

- a. Represent these data as a matrix of one hot encoded variables
- b. Compute the sample proportions for each label

Consider the data:

$$X = [1, 3, 3, 6, 11]$$

- c. Compute and sketch the empirical CDF for this sample.

Q2.

We often want to take transformations of random variables, particularly when doing feature engineering.

- a. Suppose we have a random variable X and transform it as $Y = a + bX$, with $b > 0$. The distribution of X is given by $F_X(x)$. What is the distribution of Y , $F_Y(y)$? What is the density, $f_Y(y)$?

Suppose your sample is:

$$X = [1, 3, 4]$$

- b. What is the mean of the sample?
- c. Square all the values. What the mean of X^2 ?
- d. Compute the square of the mean from part b. Does this equal the mean of the square from part c?

Non-linear transformations change the moments/statistics of the data in a non-linear way. This is important to remember.

Q3.

The **exponential distribution** is

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$

- a. Sketch a graph of the exponential distribution and density.
- b. What is the probability that $X \geq 3$? What is the probability that $X \leq 35$?
- c. Suppose we transform X by $Y = 1 + 3X$. What is the distribution of Y , $F_Y(y) = pr[Y \leq y]$? Provide a formula and sketch a graph in comparison to F_X . What is the probability that Y is less than 10?

Q4.

The **logistic distribution** is

$$F(x) = \frac{1}{1 + e^{-x}}$$

- a. Sketch a graph of the logistic distribution and density.
- b. What is the probability that $X \geq .8$? What is the probability that $X \leq .3$?
- c. Suppose we transform X by $Y = 2X - 1$. What is the distribution of Y ? Provide a formula and sketch a graph. What is the probability that Y is less than 0?

Q5.

The **median** is the value x for which the probability that X is above or below x is $.5$, or $F(\text{median}) = \frac{1}{2}$.

- a. What is the median of the exponential distribution?
- b. What is the median of the logistic distribution?

The **quantile function** is the inverse of the distribution function. The CDF answers the question, "What fraction u of the time is X below x ?" ($F(x) = u$) and the quantile function answers the question, "For what value x is X below x with probability u ?" ($F^{-1}(u) = x$)

- c. What is the quantile function of the exponential distribution?
- d. What is the quantile function of the logistic distribution?

Q6.

Load `./data/metabric.csv`.

- a. Make an ECDF plot of **Overall Survival (Months)**.
- b. Make an ECDF plot of **Overall Survival (Months)**, **hued by Chemotherapy**. Conditional on a patient receiving Chemotherapy, should we predict they will live a longer or shorter amount of time? Explain your answer clearly.
- c. Is chemotherapy an effective treatment? Explain your answer clearly.